e) Determine the solution verifying the initial conditions

$$y(0) = 0,\ y'(0) = 0\ y''(0) = 1,\ y'''(0) = 1.$$

f) Explain why it is convenient to use the functions $\cosh t$ and $\sinh t$ rather than e^t and e^{-t}.

8. Determine the general solution of the given differential equation.

a) $y''' - y'' - y' + y = 2e^{-t} + 3$

b) $y^{(4)} - y = 3t + \cos t$

c) $y''' + y'' + y' + y = e^{-t} + 4t$

d) $y^{(4)} - 4y'' = t^2 + e^t$

e) $y^{(4)} + 2y'' + y = 3 + \cos(2t)$

f) $y^{(6)} + y''' = t$

9. Find the solution of the given initial value problem.

a) $y''' + 4y' = t;\ \ y(0) = y'(0) = 0,\ y''(0) = 1$

b) $y^{(4)} + 2y'' + y = 3t + 4;\ \ y(0) = y'(0) = 0,\ y''(0) = y'''(0) = 1$

c) $y^{(4)} + 2y''' + y'' + 8y' - 12y = 12\sin t - e^{-t}\ \ y(0) = 3,\ y'(0) = 0,\ y''(0) = -1,\ y'''(0) = 2$

10. Determine a suitable form for particular solution $Y(t)$ if the method of undetermined coefficients is to be used. Do not evaluate the coefficients.

a) $y''' - 2y'' + y' = t^3 + 2e^t$

b) $y''' - y' = te^{-t} + 2\cos t$

c) $y^{(4)} - y''' - y'' + y' = t^2 + 4 + t\sin t$

d) $y^{(4)} + 2y''' + 2y'' = 3e^t + 2te^{-t} + e^{-t}\sin t$

CHAPTER 5

Power Series Review

1. Determine radius and interval of convergence for following power series.

a) $\displaystyle\sum_{n=0}^{\infty} \frac{2^{-n}}{n+1}(x-1)^n$

b) $\displaystyle\sum_{n=0}^{\infty} \frac{3^n}{n!}x^n$

c) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{n^3}(x-2)^n$

2. Find a power series series expansion of $f'(x)$, given the expansion for $f(x)$.

a) $f(x) = \sin x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}$

b) $f(x) = \sum_{n=0}^{\infty} a_n x^{2n}$

c) $f(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$

3. Express the given power series as a series with a generic term x^k.

a) $\sum_{n=1}^{\infty} n a_n x^{n-1}$

b) $\sum_{n=0}^{\infty} a_n x^{n+1}$

c) $\sum_{n=2}^{\infty} n(n-1) a_n x^{n+2}$

d) $\sum_{n=1}^{\infty} \frac{a_n}{n+3} x^{n+3}$

4. Show that

a) $x^2 \sum_{n=0}^{\infty} n(n+1) a_n x^n = \sum_{n=2}^{\infty} (n-2)(n-1) a_{n-2} x^n$

b) $2 \sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=1}^{\infty} n b_n x^{n-1} = b_1 + \sum_{n=1}^{\infty} \left(2a_{n-1} + (n+1) b_{n+1}\right) x^n$

Series Solutions around an ordinary point

5. Determine all singular points of the given differential equation.

a) $(x+1)y'' - x^2 y' + 3y = 0$

b) $x^2 y'' + 3y' - xy = 0$

c) $(\theta^2 - 2)y'' + 2y' + \sin(\theta) y = 0$

d) $(x^2 + x)y'' + 3y' - 6xy = 0$

e) $(t^2 - t - 2)y'' + (t+1)y' - (t-2)y = 0$

f) $(x^2 - 1)y'' + (1-x)y' + (x^2 - x + 1)y = 0$

g) $(\sin x)y'' + (\cos x)y = 0$

h) $e^x y'' - (x^2 - 1)y' + 2xy = 0$

i) $\sin(\theta) y'' - (\ln \theta) y = 0$

6. Find first four nonzero terms in a power series expansion about $x = 0$ for a general solution to the given differential eqaution.

a) $y' + (x+2)y = 0$

b) $z'' - x^2 z = 0$

c) $y' - y = 0$

d) $(x^2 + 1)y'' + y = 0$

e) $y'' - 2y' + y = 0$

7. Find the power series expansion about $x = 0$ for a general solution to the given differential equation. Your answer should include a general formula for the coefficients.

a) $y' - 2xy = 0$

b) $y'' - xy' + 4y = 0$

c) $y'' - x^2y' - xy = 0$

d) $(x^2 + 1)y'' - xy' + y = 0$

e) $y'' + y = 0$

f) $y'' - xy = 0$

8. Find the first four nonzero terms in a power series expansion about $x = 0$ for the solution to the given initial value problem.

a) $y'' + 3xy' - y = 0; \quad y(0) = 2, \ y'(0) = 0$

b) $(x^2 - x + 1)y'' - y' - y = 0; \quad y(0) = 0, \ y'(0) = 1$

c) $(x + 1)y'' - y = 0; \quad y(0) = 0, \ y'(0) = 1$

d) $y'' + (x - 2)y' - y = 0; \quad y(0) = -1, \ y'(0) = 0$

Series Solution Theorem

9. Find a minimum value for the radius of convergence of a power series solution about x_0.

a) $(x + 1)y'' - 3xy' + 2y = 0; \quad x_0 = 1$

b) $y'' - xy' - 3y = 0; \quad x_0 = 2$

c) $(x^2 - 5x + 6)y'' - 3xy' - y = 0; \quad x_0 = 0$

d) $(x^3 + 1)y'' - xy' + 3x^2y = 0; \quad x_0 = 1$

Euler Equation

10. Find a general solution to given Euler equation for $x > 0$.

a) $x^2y'' + 6xy' + 6y = 0$

b) $2x^2y'' + 13xy' + 15y = 0$

c) $x^2y'' - xy' + 17y = 0$

d) $x^2y'' + 2xy' - 3y = 0$

11. Solve the given initial value problem:

a) $t^2x'' - 12x = 0; \quad x(1) = 3, \ x'(1) = 5$

b) $x^2y'' + 5xy' + 4y = 0; \quad y(1) = 3, \quad y'(1) = 7$

12. Let

$$L[y](x) := x^3y''' + xy' - y.$$

a) Show that

$$L[x^r] = (r-1)^3x^r.$$

b) Using an extension of the argument given in this section for a double root, show that $L[y] = 0$ has the general solution

$$y(x) = c_1x + c_2x\ln x + c_3x(\ln x)^2.$$

Regular Singular Points, Part I

13. Classify each singular point of the given equation as regular or irregular.

a) $(x^2-1)y'' + xy' + 3y = 0$

b) $x^2y'' + 8xy' - 3xy = 0$

c) $(x^2+1)y'' + 7x^2y' - 3xy = 0$

d) $x^2y'' - 5xy' + 7y = 0$

e) $(x^2-1)^2y'' - (x-1)y' + 3y = 0$

f) $(x^2-4)y'' + (x+2)y' + 3y = 0$

g) $(x^2-x-2)y'' + (x^2-4)y' - xy = 0$

h) $(x^2-x)y'' + (x^2-3x)(\sin x)y' - xy = 0$

14. Find the indicial equation and the exponents for the specified singularity of the given differential equation.

a) $x^2y'' - 2xy' - 10y = 0,$ at $x = 0$

b) $x^2y'' + 4xy' + 2y = 0,$ at $x = 0$

c) $(x^2-x-2)^2y'' + (x^2-4)y' - 6xy = 0,$ at $x = 2$

d) $(x^2-4)y'' + (x+2)y' + 3y = 0,$ at $x = -2$

e) $x^3y'' + x(\sin x)y' - (\tan x)y = 0,$ at $x = 0$

f) $(x^2-1)y'' - (x-1)y' - 3y = 0,$ at $x = 1$

g) $(x-1)^2y'' + (x^2-1)y' - 12y = 0,$ at $x = 1$

h) $4x(\sin x)y'' - 3y = 0,$ at $x = 0$

15. Use the indicial equation to find at least one solution of the form

$$y(x) = x^r \sum_{n=0}^{\infty} a_nx^n.$$

a) $4x^2y'' + 2x^2y' - (x+3)y = 0$

b) $x^2y'' + (x^2-x)y' + y = 0$

c) $xw'' - w' - xw = 0$

d) $3x^2y'' + 8xy' + (x-2)y = 0$

e) $xy'' + (x-1)y' - 2y = 0$

f) $x(x+1)y'' + (x+5)y' - 4y = 0$

16. For each of the following problems:

i) Show that the given differential equation has a regular singular point.

ii) Determine the indicial equation, the recurrence relation, and the roots of the indicial equation.

iii) Find the series solution $(x > 0)$ corresponding to the larger root.

iv) If the roots are different and do not differ by an integer, find the series solution corresponding to the smaller root also.

a) $2xy'' + y' + xy = 0$

b) $x^2y'' + xy' + \left(x^2 - \frac{1}{9}\right)y = 0$

c) $xy'' + y = 0$

d) $xy'' + y' - y = 0$

e) $x^2y'' + xy' + (x-2)y = 0$

f) $xy'' + (1-x)y' - y = 0$

17. The Chebyshev equation is

$$(1-x^2)y'' - xy' + \alpha^2 y = 0,$$

where α is a constant.

a) Show that $x = 1$ and $x = -1$ are regular singular points, and find the exponents at each of these singularities.

b) Find two solutions about $x = 1$.

18. The Bessel equation of order is

$$x^2y'' + xy' + x^2y = 0.$$

a) Show that $x = 0$ is a regular singular point.

b) Show that the roots of the indicial equation are $r_1 = r_2 = 0$.

c) Find one solution.

d) Show that the series you found in part (c) converges for all x.

19. The Laguerre equation of order is

$$xy'' + (1-x)y' + \lambda y = 0.$$

a) Show that $x = 0$ is a regular singular point.

b) Find the indicial equation, its roots, and the recurrence relation.

c) Find one solution.

PROBLEMS

In each of Problems 1 through 12:

(a) Find all the regular singular points of the given differential equation.

(b) Determine the indicial equation and the exponents at the singularity for each regular singular point.

1. $xy'' + 2xy' + 6e^x y = 0$
2. $x^2y'' - x(2+x)y' + (2+x^2)y = 0$
3. $x(x-1)y'' + 6x^2y' + 3y = 0$
4. $y'' + 4xy' + 6y = 0$
5. $x^2y'' + 3(\sin x)y' - 2y = 0$
6. $2x(x+2)y'' + y' - xy = 0$
7. $x^2y'' + \frac{1}{2}(x + \sin x)y' + y = 0$
8. $(x+1)^2y'' + 3(x^2-1)y' + 3y = 0$
9. $x^2(1-x)y'' - (1+x)y' + 2xy = 0$
10. $(x-2)^2(x+2)y'' + 2xy' + 3(x-2)y = 0$
11. $(4-x^2)y'' + 2xy' + 3y = 0$
12. $x(x+3)^2y'' - 2(x+3)y' - xy = 0$

In each of Problems 13 through 17:

(a) Show that $x = 0$ is a regular singular point of the given differential equation.

(b) Find the exponents at the singular point $x = 0$.

(c) Find the first three nonzero terms in each of two solutions (not multiples of each other) about $x = 0$.

13. $xy'' + y' - y = 0$
14. $xy'' + 2xy' + 6e^x y = 0$; see Problem 1
15. $x(x-1)y'' + 6x^2y' + 3y = 0$; see Problem 3
16. $xy'' + y = 0$
17. $x^2y'' + (\sin x)y' - (\cos x)y = 0$
18. (a) Show that

$$(\ln x)y'' + \tfrac{1}{2}y' + y = 0$$

has a regular singular point at $x = 1$.

(b) Determine the roots of the indicial equation at $x = 1$.

(c) Determine the first three nonzero terms in the series $\sum_{n=0}^{\infty} a_n(x-1)^{r+n}$ corresponding to the larger root. Take $x - 1 > 0$.

(d) What would you expect the radius of convergence of the series to be?